EAST POINT COLLEGE OF ENGINEERING ANDTECHNOLOGY

Affiliated to Visvesvaraya Technological University (VTU), Belagavi

Project Title

AWS SageMaker Notebook – CSV Data Analysis and Visualization

Name: D VARSHA
USN: 1EP23CI010
Course: B.E CSE (AIML)
Faculty: AJEY KUMAR B K [MICROSOFT AZURE TRAINING]
Date: 07.07.26

Problem Statement

Organizations generate large volumes of employee-related data containing information such as employee IDs, names, departments, salaries, and years of experience. Analyzing this data manually is time-consuming, error-prone, and does not provide meaningful insights for decision-making. There is a need for a scalable cloud-based solution that can securely store datasets, process them efficiently, and visualize key information in an easy-to-understand format.

The objective of this project is to use Amazon Web Services (AWS), specifically Amazon S3 and Amazon SageMaker, to perform cloud-based data analysis. The employee dataset is stored in an Amazon S3 bucket, accessed through an Amazon SageMaker Jupyter Notebook, analyzed using the Pandas library, and visualized using Matplotlib. The generated charts help compare employee salaries, understand department distribution, and analyze employee experience, enabling efficient data interpretation and decision-making.

Step 1: Open AWS Console

Sign in to AWS Management Console. Search for Amazon S3 and open the service. Amazon S3 is used to store datasets that will later be accessed from SageMaker.

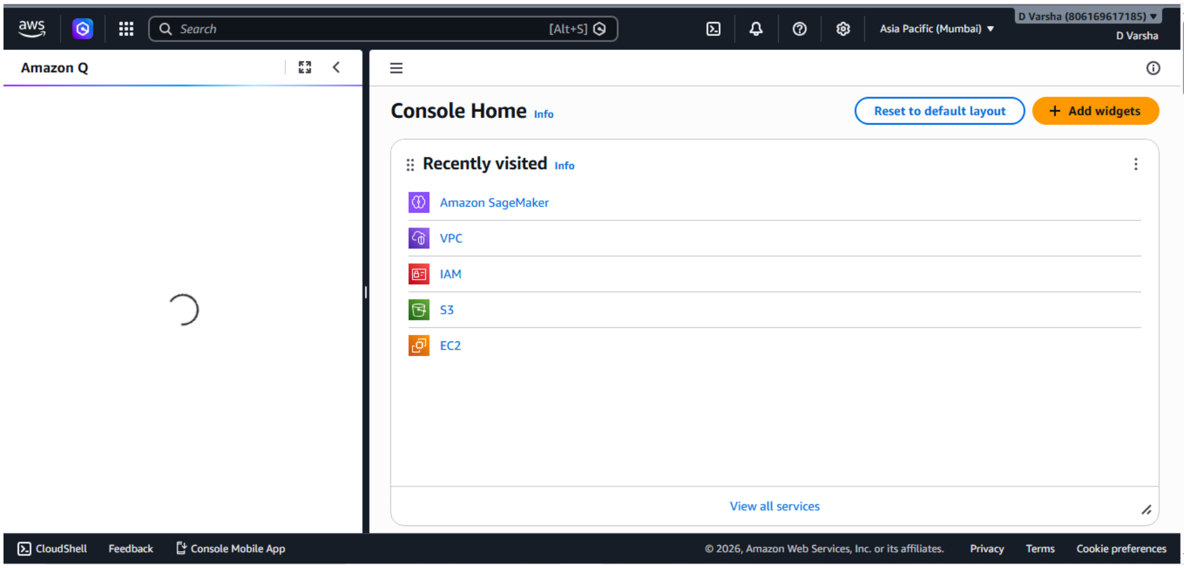

Figure 1

Step 2: Create S3 Bucket

Click Create bucket, enter a globally unique bucket name, keep default settings, and review the configuration before proceeding.

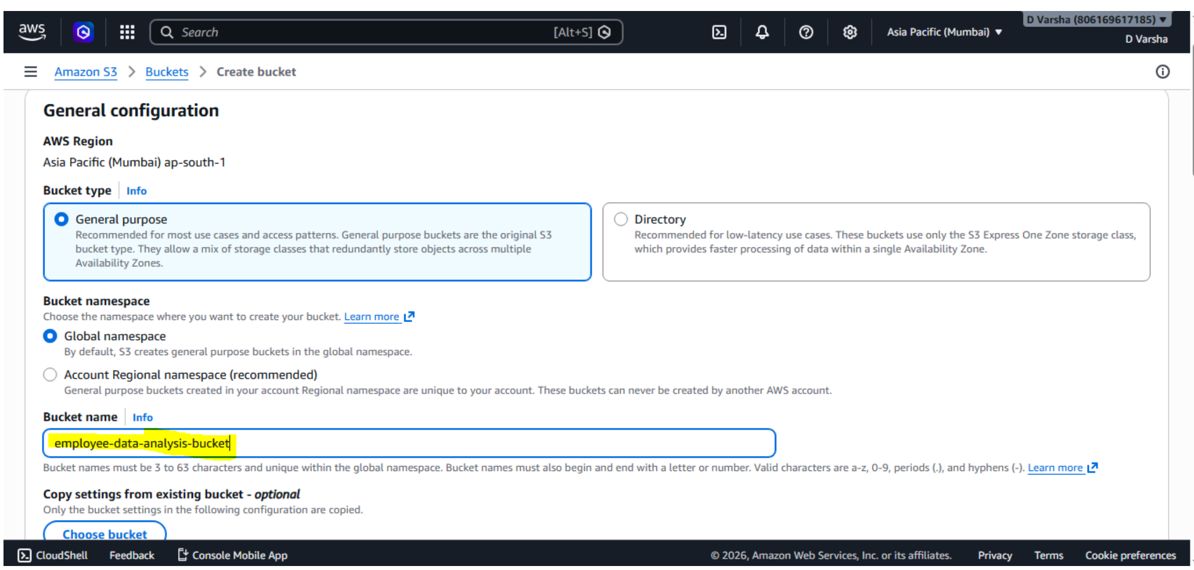

Figure 2

Step 3: Create Bucket

Scroll to the bottom of the page and click Create bucket. Wait until the bucket appears in the S3 bucket list.

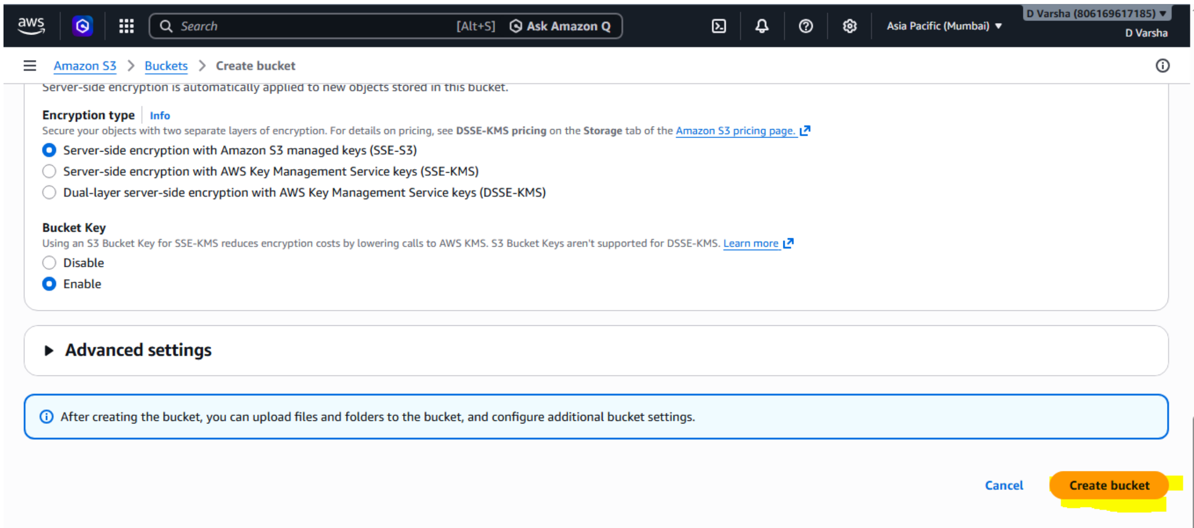

Figure 3

Step 4: Upload Dataset

Open the bucket, click Add files, select employee_data.csv from your computer, and upload it successfully.

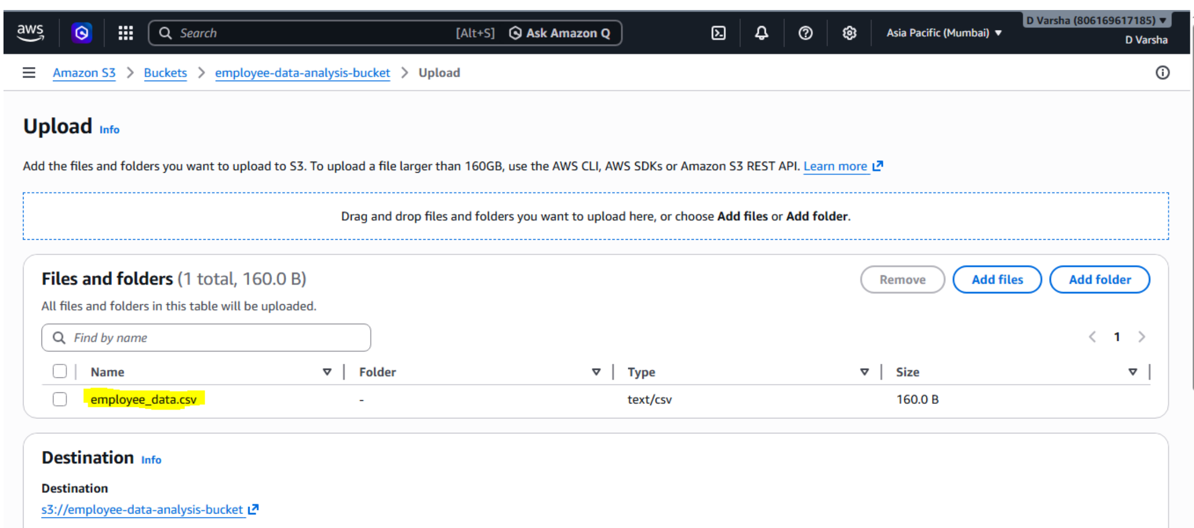

Figure 4

Step 5: Open SageMaker

Search for Amazon SageMaker AI from the AWS Console and open the service dashboard.

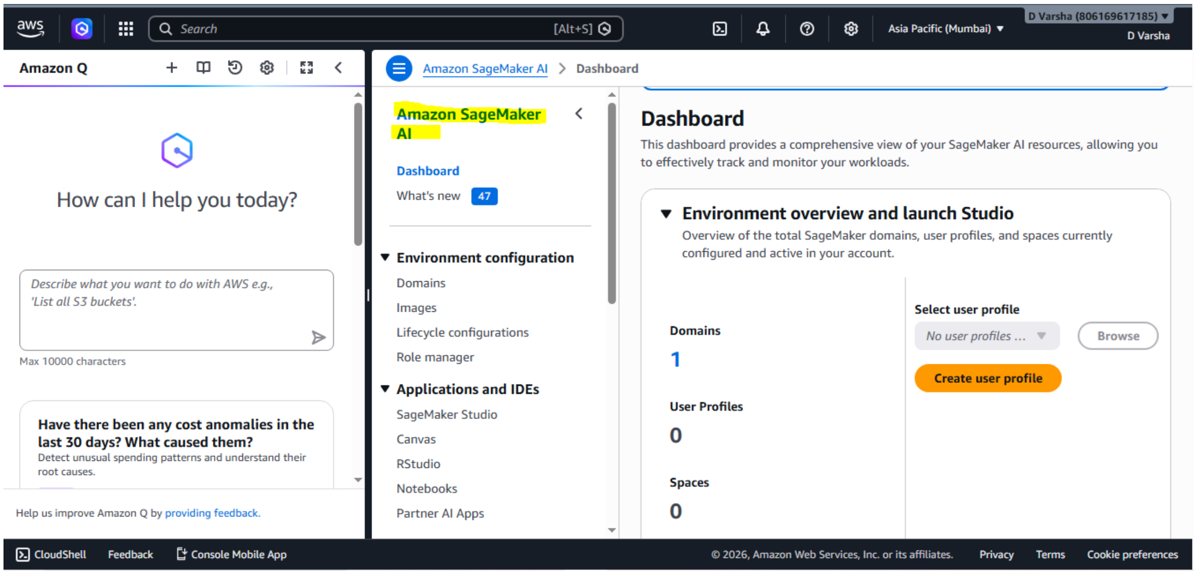

Figure 5

Step 6: Create Notebook

Navigate to Notebook Instances, click Create notebook instance, specify a notebook name, choose ml.t3.medium, and continue.

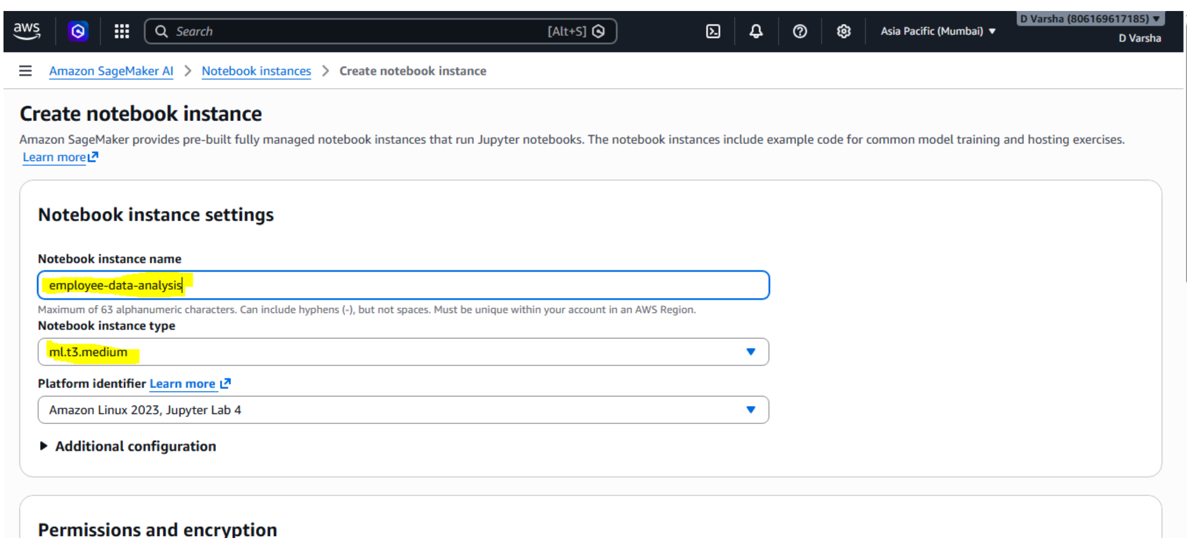

Figure 6

Step 7: Configure IAM Role

Create a new IAM role, select Any S3 bucket access, and finish the role creation so the notebook can access S3.

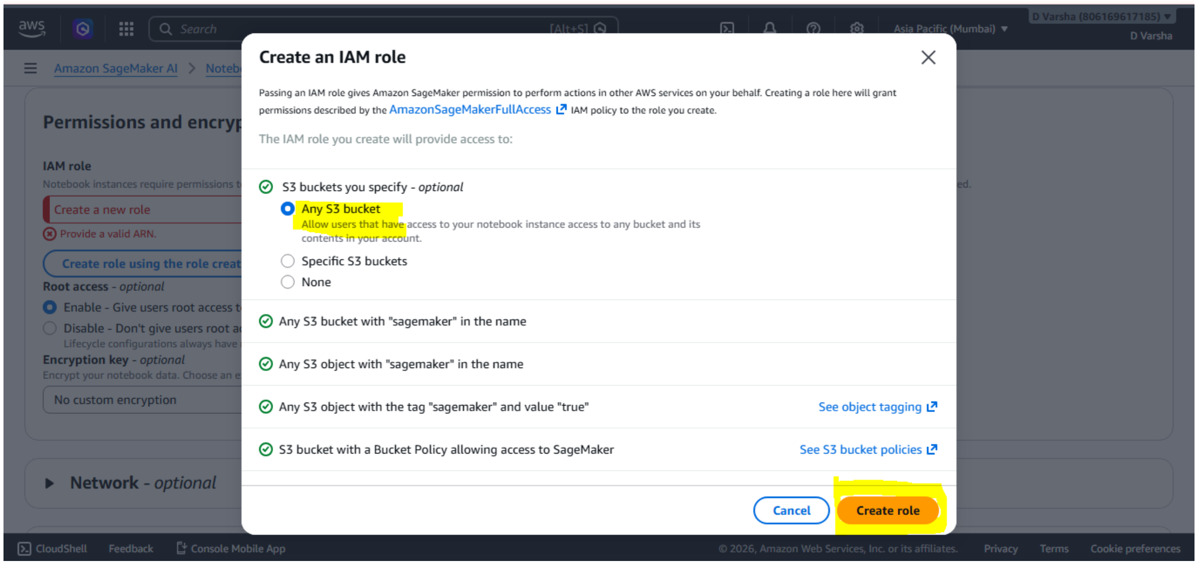

Figure 7

Step 8: Launch Notebook

Review all notebook settings and click Create notebook instance. Wait until the notebook status changes to InService.

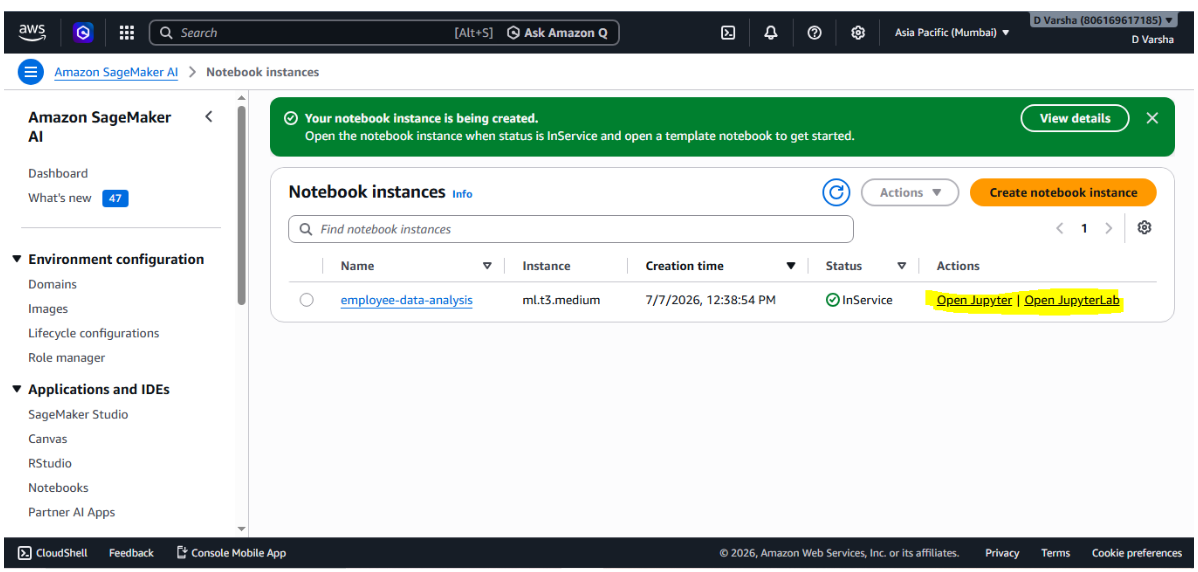

Figure 8

Step 9: Open Jupyter

Click Open Jupyter or Open JupyterLab. Create a new notebook using the conda_python3 kernel.

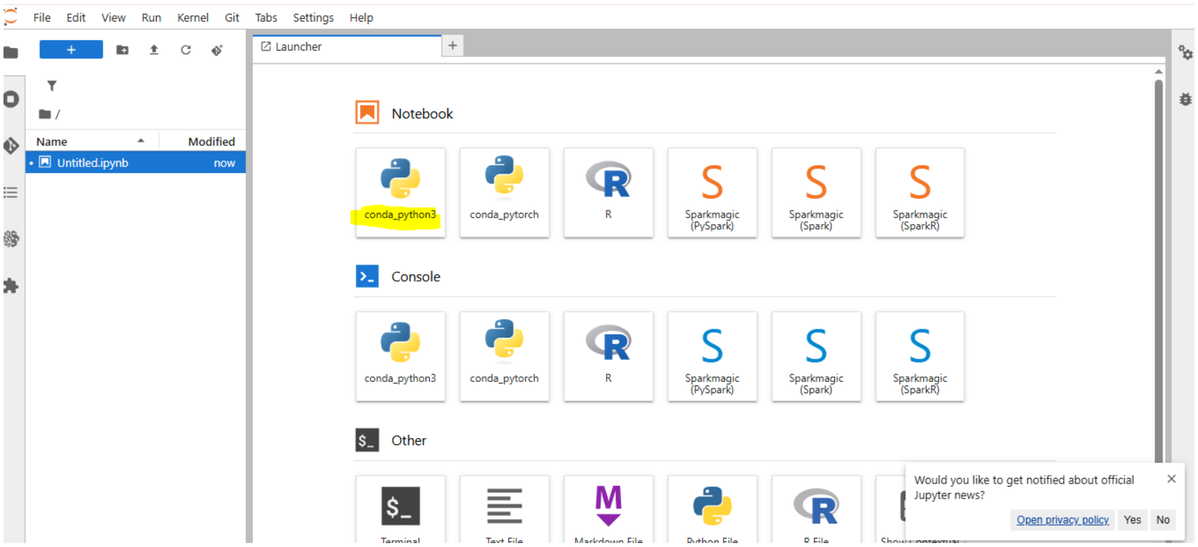

Figure 9

Step 10: Load Dataset

Import pandas, numpy, and matplotlib. Read employee_data.csv using pandas and verify the dataset using df.head().

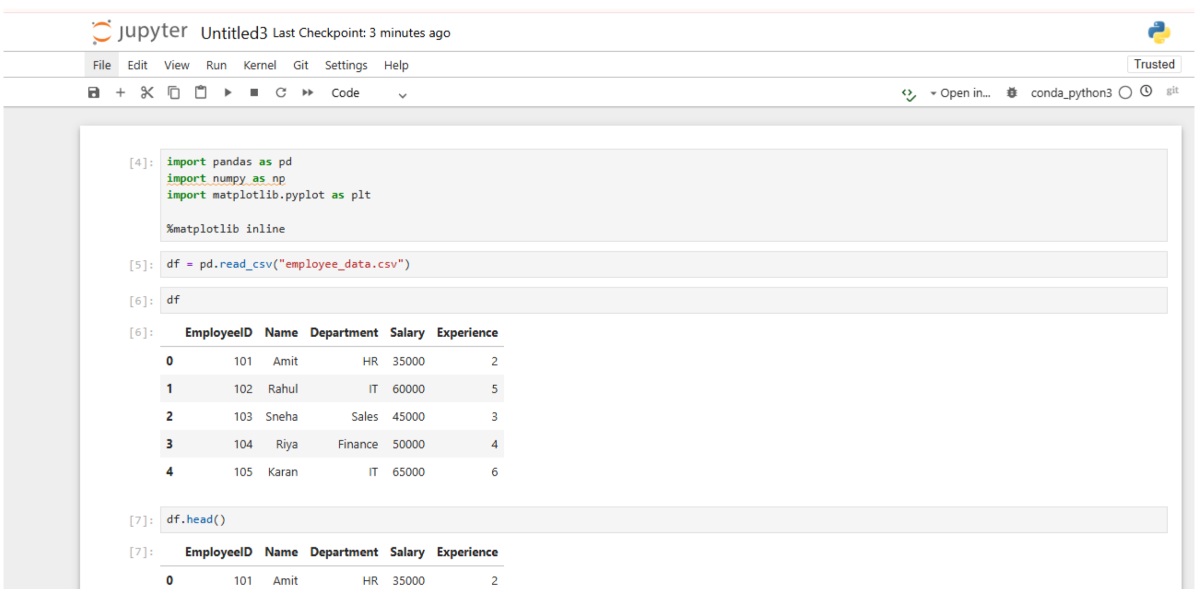

Figure 10

Step 11: Salary Bar Chart

Generate a bar chart to compare employee salaries. Add axis labels and display the plot using plt.show().

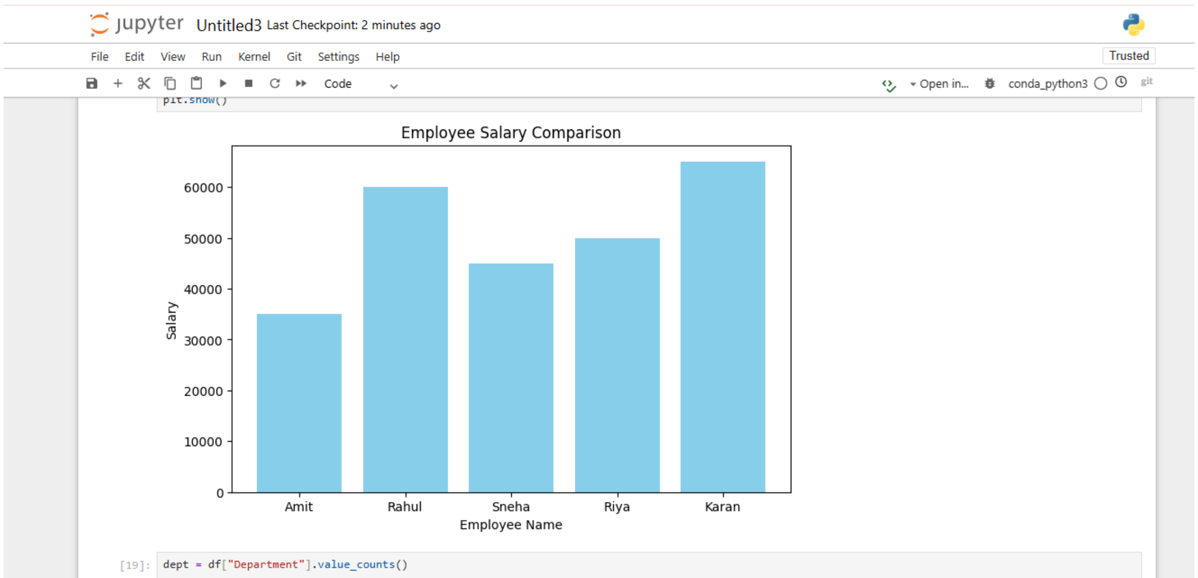

Figure 11

Step 12: Pie & Line Charts

Generate the department distribution pie chart and employee experience line chart to complete the visualization process.

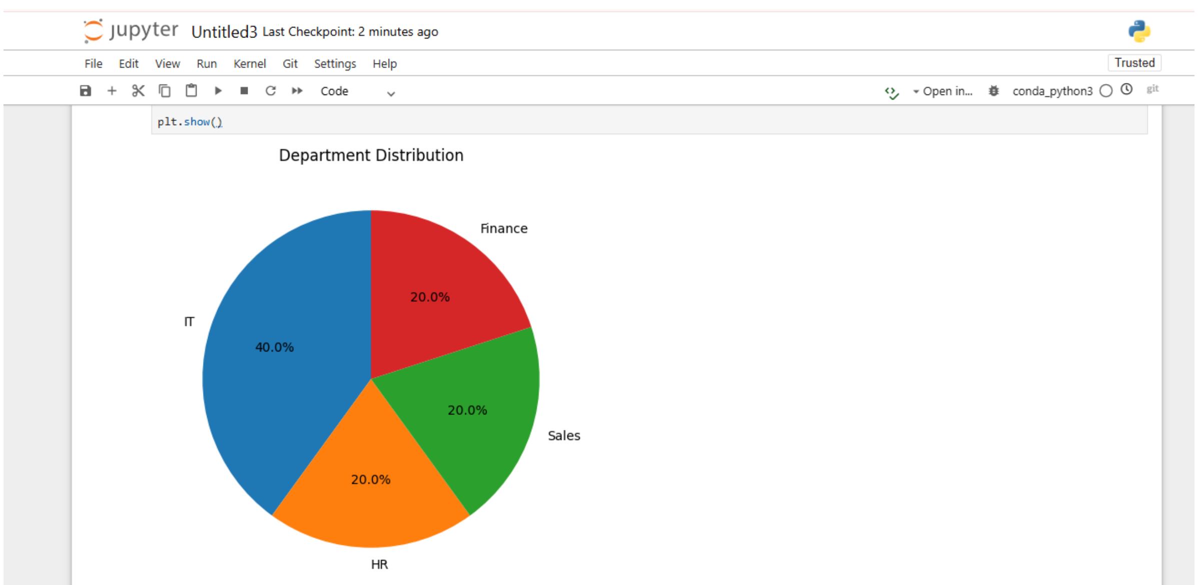

Figure 12